

Supplementary Material

Generalization-Aware Safety Monitoring for Surgical Segmentation Failures under Acquisition Degradation

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S1. Ablation and Robustness Controls

Table 1 reports the controls summarized in the main paper. They are included to bound, not expand, the main claim: confidence cues are strong, non-confidence features add modest LOCO gains, and the representation is not uniquely tied to XGBoost.

Table 1: Ablation and robustness controls. Metrics are AUROC/AUPRC. Clean feature-family ablation uses $\tau = 0.5$; remaining controls use $\tau = 0.75$.

Control	Setting	Full monitor	Comparator
Feature families	Clean test	0.593 / 0.092	Confidence-only: 0.440 / 0.024
Learned confidence-only	LOCO micro	0.920 / 0.903	Confidence-only: 0.912 / 0.888
Learner ablation	LOCO micro	XGBoost: 0.920 / 0.903	Random forest: 0.932 / 0.922
Temporal-order stress	LOCO micro	Normal: 0.920 / 0.903	Shuffled: 0.898 / 0.875
Held-sequence stress	Corrupted seq. 8–10	Combined: 0.933 / 0.928	Confidence-only: 0.894 / 0.882

S2. Calibration Burden and Held-Out Coverage

Mondrian calibration improves severity-wise miss-rate balance by increasing frame-level alarms, especially for mild degradation. Clean-test calibration is rare-event dominated, and global held-out coverage is unstable under the small calibration-positive count and temporal dependence of surgical video.

Table 2: Operational alarm burden at $\alpha = 0.10$ on corrupted severity groups.

Method	Miss-rate	False-alarm rate	Alarms/min	False alarms/min
Global	10.0%	17.6%	670.5	165.3
Mondrian	9.9%	34.2%	827.1	321.4
Severity 1, Global	50.5%	12.8%	–	–
Severity 1, Mondrian	9.3%	58.7%	–	–

Table 3: Additional conformal checks.

Setting	Global / Naive	Mondrian / Status
Clean low-quality tertile	13.9% miss	7.5% miss
Clean mid-quality tertile	5.2% miss	5.9% miss
Clean high-quality tertile	0.0% miss	8.7% miss
Cal-split threshold on held-out test	70.2% miss	Fails
Bootstrap cal/test splits ($n = 200$)	9.1% $\pm$ 5.8% miss	Uncertified

S3. SAM2 Transfer Details and Seed Stability

The U-Net-trained monitor transfers to SAM2 despite a large morphology shift: SAM2 produces many more connected components and larger perimeters. Entropy is stronger on SAM2 because zero-shot failures are expressed as uncertainty.

S4. External-Audit Caveat

A cached CholecSeg8k audit used 8080 frames, but the current external prediction/label pipeline produced failures for every frame at $\tau = 0.5, 0.6$, and 0.75 (mean IoU 0.128). AUROC and AUPRC are therefore unidentifiable on that audit, so it is not reported as external validation.

S5. Reviewer-Concern Mapping

Table 4: SAM2 details and seed stability.

Check	Value 1	Value 2	Value 3
U-Net fail rate	$\tau = 0.5$: 1.7%	$\tau = 0.75$: 6.9%	–
SAM2 fail rate	$\tau = 0.5$: 45.5%	$\tau = 0.75$: 51.2%	–
Feature shift	components: 2.5→33.2	perimeter: 1121→2734	fg prob: 0.984→0.831
Seed 0 clean AUROC	$\tau = 0.5$: 0.877	$\tau = 0.75$: 0.793	–
Seed 1 clean AUROC	$\tau = 0.5$: 0.895	$\tau = 0.75$: 0.803	–
Seed 2 clean AUROC	$\tau = 0.5$: 0.872	$\tau = 0.75$: 0.714	–
Mean±std	$\tau = 0.5$: 0.881±0.010	$\tau = 0.75$: 0.770±0.040	–

Table 5: How supplementary evidence maps to likely reviewer concerns.

Concern	Evidence in supplement
Monitor only learns confidence	Learned confidence-only control; full features add positive LOCO gains.
Result is classifier-specific	Random forest matches or exceeds XGBoost under LOCO.
Temporal features leak sequence order	Reset/shuffled temporal controls remain below normal temporal features under LOCO.
Safety calibration hides alarm burden	Alarm/min and false-alarm/min are reported explicitly.
External validity is overstated	CholecSeg8k audit is disclosed as unidentifiable, not claimed as validation.